

ALI RAZA

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI Engineer specializing in RAG and Agentic AI systems, with hands-on, production-level experience across the full ML lifecycle from EDA and classical ML/DL to Computer Vision (OpenCV), NLP (Transformers), LLMs, and Generative AI. Built and deployed production-grade RAG pipelines using LangChain and ChromaDB, along with agentic AI systems using LangGraph and MCP (Model Context Protocol), including multi-agent workflows, human-in-the-loop patterns, and persistent memory. Skilled in Python, PyTorch, TensorFlow, and Scikit-learn, with strong MLOps practices API design (FastAPI), containerization (Docker), CI/CD, and cloud deployment (AWS EC2). All projects maintained and version controlled on GitHub.

EDUCATION

BS Computer Science, University of Education, Lahore .

FEATURED PROJECTS

Crypto Price Forecasting — MLOps Pipeline

Engineered a production MLOps pipeline forecasting next-hour BTC return from live Binance data. Automated weekly retraining is drift-gated (PSI/KS) with champion-challenger promotion and zero-downtime model hot-swap; full CI/CD (pytest → Docker → SSH deploy) serving a live FastAPI API on AWS EC2, tracked with MLflow (DagsHub).

Tech Stack: *Python, XGBoost, MLflow (DagsHub), FastAPI, Docker, AWS EC2, GitHub Actions, Next.js*

[Live Demo](#) | [GitHub](#)

Transaction Anomaly & Behavioral Segmentation

Built an unsupervised ML system on the PaySim dataset (6.3M+ transactions) combining an Isolation Forest, Local Outlier Factor, and Autoencoder ensemble for real-time anomaly scoring, alongside MiniBatchKMeans/GMM/DBSCAN clustering with UMAP for behavioral segmentation. Served via a Dockerized FastAPI API on AWS EC2 with automated CI/CD (pytest → Docker → SSH deploy) and MLflow tracking.

Tech Stack: *Python, Scikit-learn, MLflow, FastAPI, Docker, AWS EC2, GitHub Actions, Next.js*

[Live Demo](#) | [GitHub](#)

Insurance Premium Predictor

Built an end-to-end MLOps system predicting annual medical insurance charges, benchmarking Linear/Ridge/Lasso against a selected XGBoost regressor ($R^2 = 0.898$, $MAE \approx \$2,520$). Engineered a smoker × BMI interaction feature after EDA identified smoking as the dominant cost driver. Full lifecycle: MLflow tracking, 10 automated pytest tests, Docker, CI/CD, and live AWS EC2 deployment.

Tech Stack: *FastAPI, XGBoost, MLflow, Docker, AWS EC2, Next.js*

PPE Safety Detection

Real-time helmet compliance detection using a fine-tuned YOLOv8 model — supports both image upload and live camera feed with bounding-box overlays and confidence scores. Full MLOps pipeline: Dockerized FastAPI backend on AWS EC2 with CI/CD (GitHub Actions), DVC-tracked model versioning, and a Next.js frontend on Vercel with a mixed-content-safe server-side proxy architecture.

Tech Stack: YOLOv8 , FastAPI , Docker , AWS EC2 , GitHub Actions Next.js TypeScript

[Live Demo](#) | [Github](#)

CORE COMPETENCIES

Python, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, MLOps, Transformers, Generative Ai , Agentic Ai, Model Deployment, Prompt Engineering, FastAPI, Docker, GitHub Actions, LeetCode

TECHNICAL SKILLS

LANGUAGES: Advanced Python, C++

EDA: NumPy, Pandas, Matplotlib, Seaborn, Data Cleaning

Math for ML: Linear Algebra (Vectors, Matrices, Tensors), Calculus (Derivatives, Gradient Descent, Chain Rule)

ML: Supervised, Unsupervised & Reinforcement Learning, Feature Engineering, Feature Scaling, Scikit-learn

MLOps: MLflow (Experiment Tracking, Model Registry), DVC (Data Version Control), DagsHub

DL: ANN, CNN, RNN, LSTM, GRU, TensorFlow, Keras, PyTorch, Transfer Learning

Computer Vision: OpenCV, Image Processing, Image Classification, Object Detection, Image Augmentation, CNN, YOLO

NLP: Tokenization, Stop Words Removal, Stemming & Lemmatization, POS Tagging, NER, TF-IDF, Word Embeddings, Sentiment Analysis

Generative AI / LLMs: LLMs, Vector Embeddings, ChromaDB, RAG, LangChain, Prompt Engineering, Groq, OpenAI

Agentic AI: LangGraph, Checkpointers (SQLite/Postgres), Human-in-the-Loop, Multi-Agent Workflows, MCP, LangSmith

Backend: Python, FastAPI, PostgreSQL, SQLite, Async Programming, API Authentication **Deployment:**

Docker, AWS EC2, GitHub Actions (CI/CD), Vercel, Containerization

SOFT SKILLS

- Analytical Thinking
- Problem Solving
- Communication
- Collaboration,
- Technical Documentation
- Adaptability
- Time Management
- Research

LANGUAGES

English (Professional Working Proficiency)
Urdu (Native)